

Arduino-Based Obstacle Detection Using Ultrasonic Sensor and Buzzer

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Abstract—This paper presents the design and simulation of an obstacle detection system using Arduino Uno, HC-SR04 ultrasonic sensor, and a piezo buzzer. The system measures distance to nearby objects and triggers an audible alert when the object is within a safety threshold. The prototype was implemented in Tinkercad Circuits, providing a low-cost simulation environment without requiring physical hardware.

Index Terms—Arduino Uno, Ultrasonic Sensor, HC-SR04, Obstacle Detection, Simulation, Tinkercad

I. INTRODUCTION

Obstacle detection is a fundamental requirement in robotics and automation, enabling systems to sense and respond to their environment. Ultrasonic sensors are widely used for distance measurement due to their low cost and reliability. This project implements a low-cost solution using an HC-SR04 ultrasonic sensor to measure distance and a buzzer to provide feedback. The aim was to design and simulate a working prototype within two days using only Tinkercad Circuits.

II. METHODOLOGY

A. Hardware (Simulated)

Arduino Uno R3, HC-SR04 ultrasonic sensor, piezo buzzer, breadboard, and jumper wires.

B. Software

Arduino IDE (C/C++ code) and Tinkercad Circuits simulation platform.

C. Circuit Design and Wiring

The TRIG pin of the HC-SR04 sensor was connected to Arduino digital pin 9, while the ECHO pin was connected to digital pin 10. The buzzer was connected to digital pin 8 through a series resistor to limit current. Proper 5V and GND connections were ensured for both the sensor and buzzer to maintain stable operation.

D. Simulation Setup

Tinkercad Circuits was used to simulate breadboard wiring and virtual testing. Distance values were manually adjusted in the HC-SR04 sensor properties panel to emulate real-world scenarios. The Serial Monitor was enabled to observe real-time distance readings during simulation.

E. Algorithm

- 1) Trigger ultrasonic pulse via TRIG pin.
- 2) Measure echo duration via ECHO pin.
- 3) Convert duration to distance in centimeters.
- 4) If distance < threshold (120 cm), activate buzzer.
- 5) Else, buzzer remains off.

F. Testing Procedure

Multiple distance values were tested (10 cm, 50 cm, 100 cm, 150 cm). Buzzer activation was observed consistently below the threshold of 120 cm. A delay of 100 ms was introduced between measurements to ensure stable readings.

G. Safety and Limitations

Simulation avoids hardware risks such as incorrect wiring or sensor damage. The HC-SR04 sensor has a practical range of approximately 400 cm in real hardware. The buzzer provides only audible feedback; no visual indicators were included in this prototype.

III. CIRCUIT SCHEMATIC

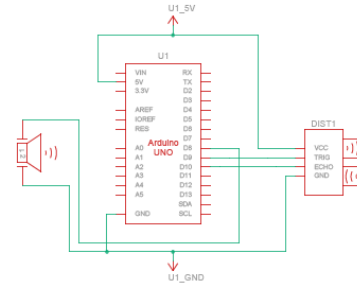


Fig. 1. Circuit diagram made using Tinkercad

IV. RESULTS

The system successfully detected obstacles at varying distances. Serial Monitor displayed real-time distance values. Buzzer activation occurred consistently when objects were within the safety threshold. Debugging revealed wiring errors on the breadboard as the primary challenge, which was solved.

V. DISCUSSION

The project highlights the importance of careful wiring and simulation setup in embedded systems. While simple, the design demonstrates core concepts of sensor integration, threshold logic, and actuator control. This approach can be extended to robotics, smart walking sticks, or automated safety systems.

VI. CONCLUSION

A functional obstacle detection system was implemented and tested entirely in simulation. The project provided hands-on experience with Arduino programming, sensor interfacing, and debugging. Future work may involve integrating multiple sensors or adding visual indicators.

REFERENCES

- [1] M. A. Baballe, "Obstacle Avoidance Robot using an Ultrasonic Sensor with Arduino Uno," *Global Journal of Research in Engineering & Computer Sciences*, 2023. DOI: 10.5281/ZENODO.10015177.
- [2] O. Mishra, "Autonomous Mobile Robot with Arduino Uno and HC-SR04," *GitHub Repository*, 2022. Available: <https://github.com/Om-Prakash-Mishra/Obstacle-Avoiding-Robot>
- [3] K. A. Raza, W. Monnet, "Moving Objects Detection and Direction-Finding with HC-SR04 Ultrasonic Linear Array," *Proc. IEC Conference*, 2019. DOI: 10.1109/IEC47844.2019.8950639.
- [4] SparkFun Electronics, "HC-SR04 Ultrasonic Ranging Module Datasheet", *Technical Documentation*, 2025. Available: <https://www.sparkfun.com>
- [5] U. G. Kulkarni, *TinkerCAD Circuits Reference Handbook*, Google Books, 2025. ISBN: 9789358134567.